“Fake Science” and Information Overload in Academia

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Information Overload (IOL)

- *"That situation which arises when there is so much relevant and potentially useful information available that it becomes a hindrance rather than a help"* (Bawden et. al)
- Link between *too much information* and our *limited human cognitive capacity*
- Negative psychological state in which people feel they receive too much information to carry out their tasks

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Janusz A. Hołyst, Philipp Mayr, Michael Thelwall, Ingo Frommholz, et al. "Protect Our Environment from Information Overload". In: *Nature Human Behaviour* (Feb. 7, 2024), pp. 1–2. ISSN: 2397-3374. DOI: 10.1038/s41562-024-01833-8.

David Bawden and Lyn Robinson. "Information Overload: An Introduction". In: *Oxford Research Encyclopedia of Politics*. June 2020. ISBN: 978-0-19-022863-7. DOI: 10.1093/acrefore/9780190228637.013.1360.

Information Overload (IOL)

- Consequences of IOL
  - Limits our ability to evaluate information
  - Decreases decision performance, increases tolerance of errors
  - Reduces social activities, lowers job and user satisfaction
  - Can lead to demotivation, negative emotions and compromised health
  - Large economic impact (estimated global costs $\sim$ US$1 trillion/year)
- IOL as pollution – the information space should be treated as part of our environment, to be protected like air, water and soil
- Information Retrieval as one remedy of IOL

Janusz A. Hołyst, Philipp Mayr, Michael Thelwall, Ingo Frommholz, et al. “Protect Our Environment from Information Overload”. In: *Nature Human Behaviour* (Feb. 7, 2024), pp. 1–2. ISSN: 2397-3374. DOI: 10.1038/s41562-024-01833-8.

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Multilevel AI Overload

Type	Definition	I	O	S
Cognitive	Working memory and attention exceed capacity; increased error rates, shallow processing, reduced deliberation.	+	-	-
Information	Excess volume/velocity/variety; low signal-to-noise; high verification burden.	+	+	+
Interaction	Too many AI touchpoints (copilots, alerts, assistants), producing attention fragmentation and frequent context switching.	+	+	-
Coordination	Humans manage multiple agents and handoffs; agent–agent conflicts; escalation and arbitration load.	+	+	+
Control /monitoring	Supervisory burden rises: audit trails, provenance checks, safety constraints; "human-in-the-loop" fatigue.	+	+	+
Normative	Uncertainty about acceptable use (ethics, compliance, intellectual property); hesitation and friction; policy churn.	+	+	+
Affective	Anxiety, learned helplessness, diminished self-efficacy, social comparison vs. AI; emotional exhaustion.	+	+	+

Przemysław Kazienko, Wiktoria Mieleszczenko-Kowszewicz, Jan Kocon, Beata Bajcar, Julian Sienkiewicz, Janusz Hołyst, Philipp Mayr, Ingo Frommholz, and Erik Cambria. "AI Overload: A Multi-Level Taxonomy and the Path Forward". In: *IEEE Intelligent Systems* 41.2 (2026).

Information Overload in Academia – Quality vs. Quantity

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Information Overload in Academia

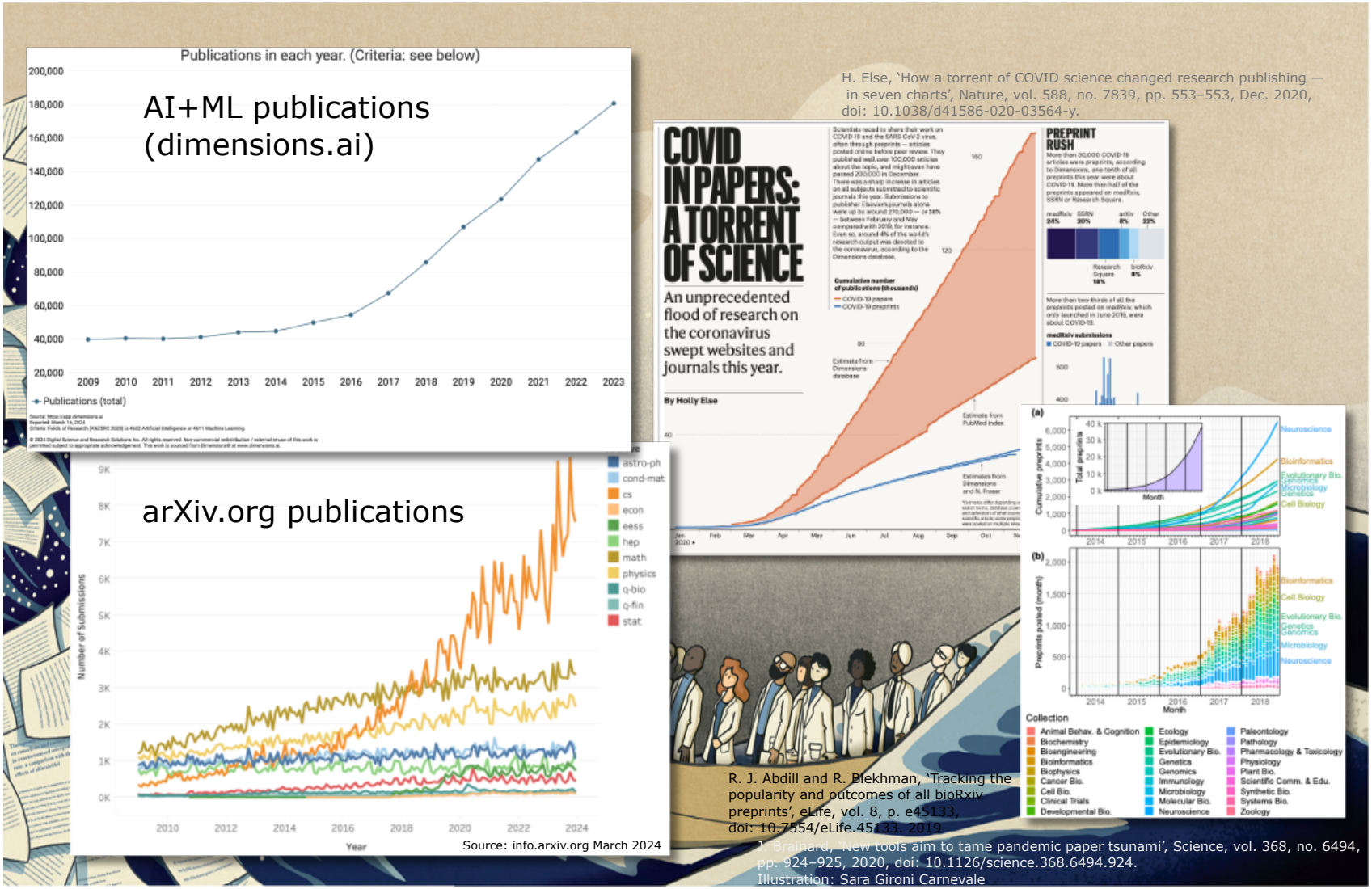
Finding relevant research in a “publication tsunami” [8, 2, 14]



J. Brainard, ‘New tools aim to tame pandemic paper tsunami’, Science, vol. 368, no. 6494, pp. 924–925, 2020, doi: 10.1126/science.368.6494.924.
Illustration: Sara Gironi Carnevale

Information Overload in Academia

Finding relevant research in a “publication tsunami” [8, 2, 14]



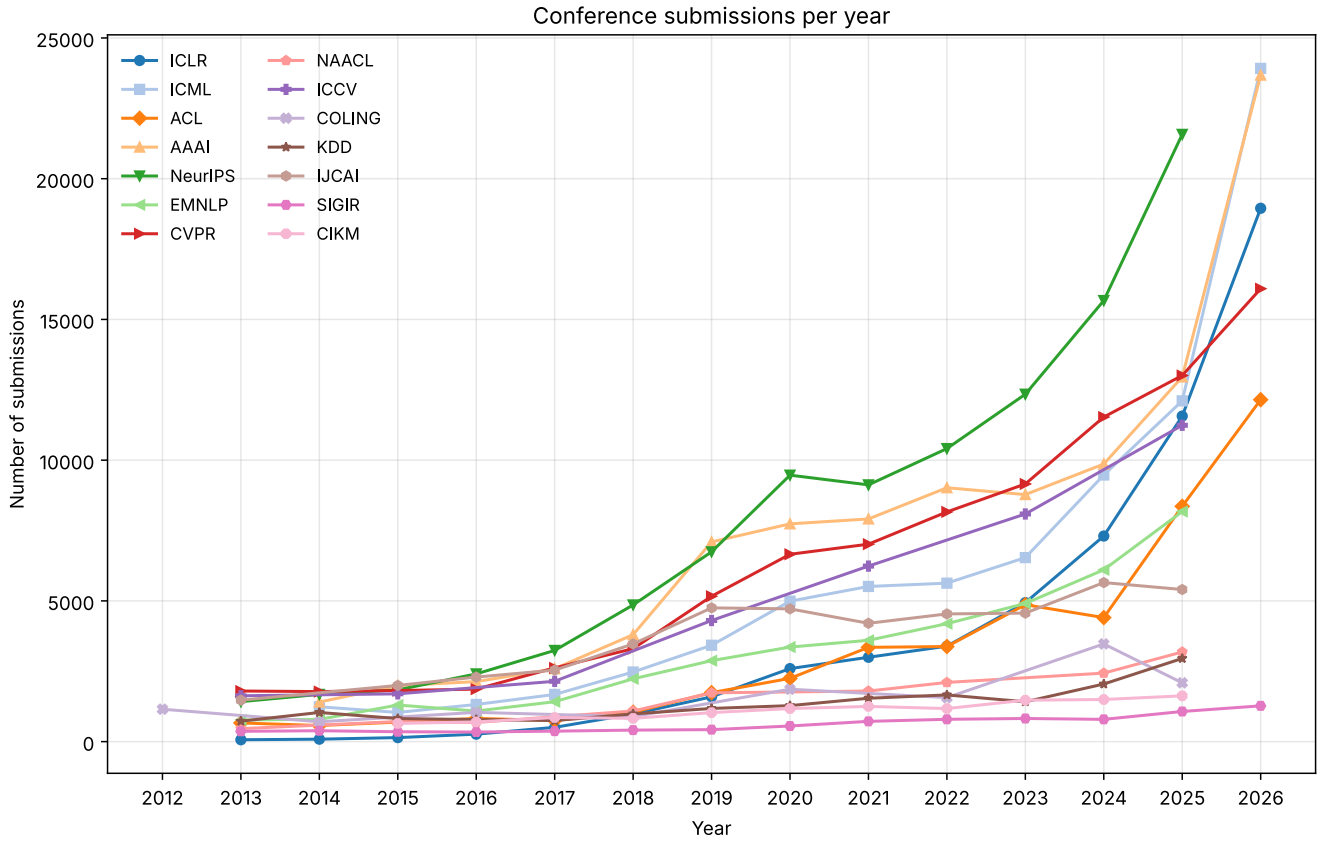
H. Else, ‘How a torrent of COVID science changed research publishing — in seven charts’, Nature, vol. 588, no. 7839, pp. 553–553, Dec. 2020, doi: 10.1038/d41586-020-03564-y.

R. J. Abdill and R. Blekhman, ‘Tracking the popularity and outcomes of all bioRxiv preprints’, eLife, vol. 8, p. e45133, doi: 10.7554/eLife.45133. 2019

J. Brainard, ‘New tools aim to tame pandemic paper tsunami’, Science, vol. 368, no. 6494, pp. 924–925, 2020, doi: 10.1126/science.368.6494.924.

Illustration: Sara Gironi Carnevale

Conference Submissions per Year



Stable acceptance rates. ICLR 2027 reportedly received 62,288 abstract submissions!

https://reviewer3.com/evidence/iclr-submissions-2027

Kenneth Ward Church. “Emerging Trends: Too Many Submissions and Too Few Qualified Reviewers”. In: *Natural Language Processing* (Sept. 2026), pp. 1–13. ISSN: 2977-0424. DOI: 10.1017/nlp.2026.10038.

arXiv CS not Accepting Survey Papers Anymore

- arXiv has been flooded with papers, especially papers not introducing new research results
- arXiv receives hundreds of review articles every month
- Survey papers are fast and easy to write – or to generate with LLMs
- Majority of the review articles are little more than annotated bibliographies, with no substantial discussion of open research issues
- Moderators are overwhelmed
- Now survey papers must have been accepted by journals or conferences, thus being peer-reviewed

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Kat Boboris. *Attention Authors: Updated Practice for Review Articles and Position Papers in arXiv CS Category – arXiv Blog*. https://blog.arxiv.org/2025/10/31/attention-authors-updated-practice-for-review-articles-and-position-papers-in-arxiv-cs-category/. Oct. 2025. (Visited on 07/12/2025).

Low-quality Publications and Information Overload

An example case

- A dramatic increase in low-quality papers that use the National Health and Nutrition Examination Survey (NHANES) data set
  - 4,876 papers in 2023 rising to 7,876 by 2024
  - Papers studying single/simple associations (PubMed and Scopus): Average 4 between 2014 and 2022, then rapid increase with 190 published in 2024
  - Timing coincides with the availability of ChatGPT
  - Only 13 out of 28 NHANES studies that examined depression survived a statistical adjustment designed to control for the risk of false discoveries

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Catherine O'Grady. "Low-Quality Papers Are Surging by Exploiting Public Data Sets and AI". In: *Science* 388 (6749 May 22, 2025). DOI: 10.1126/science.adz1715.

Information Overload, Paper Volume, Quality and Fraud

- Indication that IOL is caused partly by the volume and quality of publications [31]
- Low-quality and even fraudulent papers contribute to this “publication tsunami” [8]
- Not only preprint servers are affected – publishers retract papers too [22]
- “Industrialisation” of low-quality research: paper mill activity, pressure for rapid publication (quantity before quality)
- Finding these cases is therefore a critical task
- Challenges include
  - Tortured phrases
  - Auto-generated papers
  - Paper mills
  - Fake science

Tortured Phrases

Phrases incorrectly used instead of correct ones (e.g. to avoid plagiarism detection)

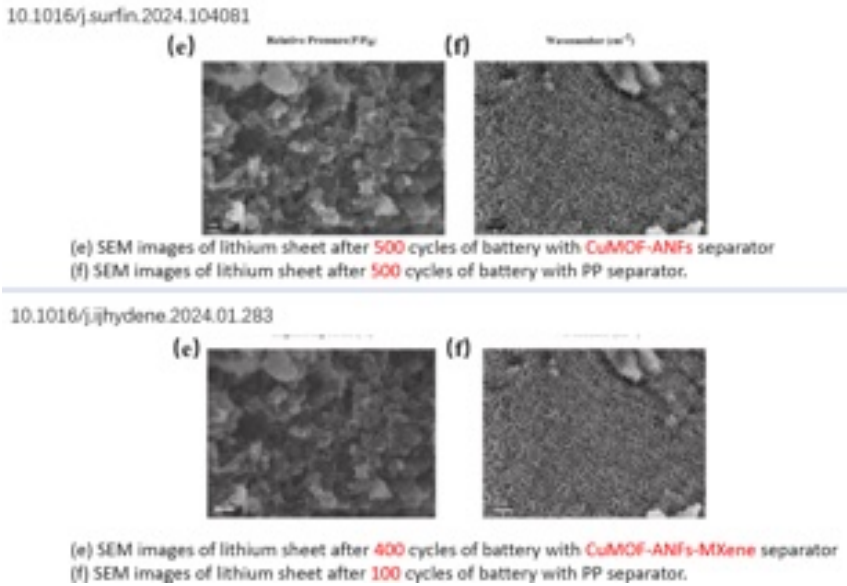
Table 1 Tortured phrases we found in the literature along with their usual, correct wording.

Tortured phrase found in publications	Correct wording expected
profound neural organization	deep neural network
(fake | counterfeit) neural organization	artificial neural network
versatile organization	mobile network
organization (ambush | assault)	network attack
organization association	network connection
(enormous | huge | immense | colossal) information	big data
information (stockroom | distribution center)	data warehouse
(counterfeit | human-made) consciousness	artificial intelligence (AI)
elite figuring	high performance computing
haze figuring	fog/mist/cloud computing

Guillaume Cabanac, Cyril Labbé, and Alexander Magazinov. *Tortured Phrases: A Dubious Writing Style Emerging in Science. Evidence of Critical Issues Affecting Established Journals*. July 2021. DOI: 10.48550/arXiv.2107.06751. arXiv: 2107.06751 [cs].

Fraudulent Papers

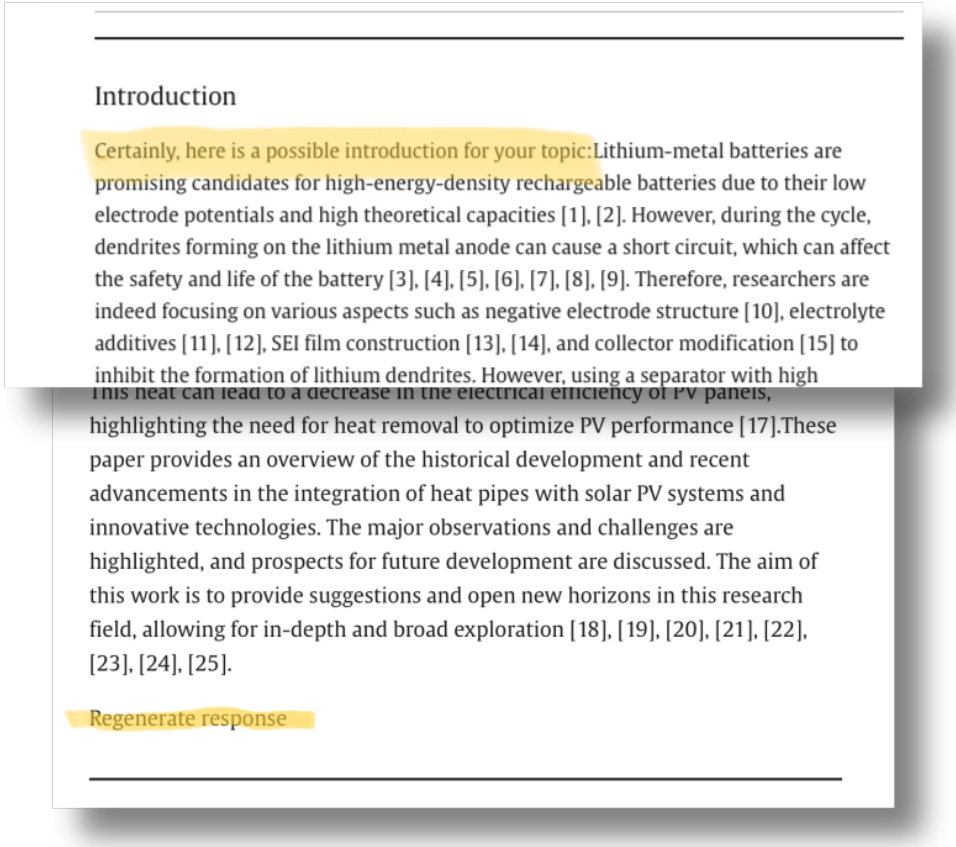
- Elsevier “Surfaces and Interfaces”, March 2024 (retracted May 2024)
- vs. Elsevier “International Journal of Hydrogen Energy”, March 2024 (not retracted as of Jan 2026)
- Same figures reused across multiple papers, claimed to represent different substances



Ateleopus Purpureus. *Comment on PubPeer – The three-dimensional porous mesh structure of Cu-based metal-organic-framework - aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries*. Mar. 2024. URL: https://pubpeer.com/publications/CAABBF887348FB2D1C0329E0A27BE6#1 (visited on 02/04/2026).

AI-generated Research Papers

- Elsevier “Surfaces and Interfaces”, March 2024 (retracted May 2024)
- Auto-generated papers? Or just a little writing aid?
- Quality of research papers?



Introduction

Certainly, here is a possible introduction for your topic:Lithium-metal batteries are promising candidates for high-energy-density rechargeable batteries due to their low electrode potentials and high theoretical capacities [1], [2]. However, during the cycle, dendrites forming on the lithium metal anode can cause a short circuit, which can affect the safety and life of the battery [3], [4], [5], [6], [7], [8], [9]. Therefore, researchers are indeed focusing on various aspects such as negative electrode structure [10], electrolyte additives [11], [12], SEI film construction [13], [14], and collector modification [15] to inhibit the formation of lithium dendrites. However, using a separator with high

This heat can lead to a decrease in the electrical efficiency of PV panels, highlighting the need for heat removal to optimize PV performance [17].These paper provides an overview of the historical development and recent advancements in the integration of heat pipes with solar PV systems and innovative technologies. The major observations and challenges are highlighted, and prospects for future development are discussed. The aim of this work is to provide suggestions and open new horizons in this research field, allowing for in-depth and broad exploration [18], [19], [20], [21], [22], [23], [24], [25].

Regenerate response

Guillaume Cabanac. *Comment on PubPeer – The three-dimensional porous mesh structure of Cu-based metal-organic-framework - aramid cellulose separator enhances the electrochemical performance of lithium metal anode batteries*. Mar. 2024. URL: https://pubpeer.com/publications/CAABBF887348FB2D1C0329E0A27BE6#1 (visited on 02/04/2026).

Paper Mills

Industrialised fraud in scholarly publishing



- Industrial-scale production of manuscripts and authorship slots
- Fabricated data, template text, and manipulated peer review
- Contributes to academic information overload
- Erodes trust in scholarly communication

Richard Van Noorden. "How Big Is Science's Fake-Paper Problem?" In: *Nature* 623.7987 (Nov. 2023), pp. 466–467. DOI: 10.1038/d41586-023-03464-x.

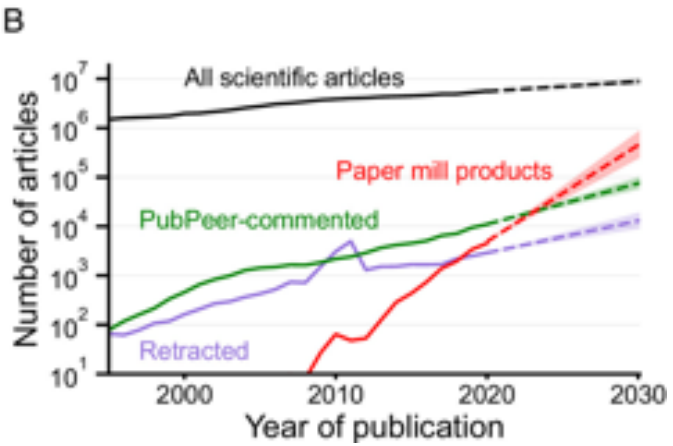
Ahmar K Hussain, Marcus Thiel, Bernhard A Sabel, and Andreas Nürnberger. *Fake Papers in Science: Paper Mills, Characteristics and Detection Strategies*.

Matt Spick, Anthony Onoja, Charlie Harrison, Stefan Stender, Jennifer Byrne, and Nophar Geifman. *Quantifying New Threats to Health and Biomedical Literature Integrity from Rapidly Scaled Publications and Problematic Research*. July 2025. DOI: 10.1101/2025.07.07.25331008.

Reese A. K. Richardson, Spencer S. Hong, Jennifer A. Byrne, Thomas Stoeger, and Luís A. Nunes Amaral. "The Entities Enabling Scientific Fraud at Scale Are Large, Resilient, and Growing Rapidly". In: *Proceedings of the National Academy of Sciences* 122.32 (Aug. 2025), e2420092122. DOI: 10.1073/pnas.2420092122.

From Paper Mills to Systemic Fake Science

- Growth Trend: Paper Mill Products > PubPeer – Commented > Retracted
- Large-scale analysis (1995–2014):
  - 3.8% of ∼20,000 articles contained duplicated images
  - ∼50% suggest deliberate manipulation



Reese A. K. Richardson, Spencer S. Hong, Jennifer A. Byrne, Thomas Stoeger, and Luís A. Nunes Amaral. "The Entities Enabling Scientific Fraud at Scale Are Large, Resilient, and Growing Rapidly". In: *Proceedings of the National Academy of Sciences* 122.32 (Aug. 2025), e2420092122. DOI: 10.1073/pnas.2420092122.

Elisabeth M Bik, Arturo Casadevall, and Ferric C Fang. "The prevalence of inappropriate image duplication in biomedical research publications". In: *MBio* 7.3 (2016), pp. 10–1128.

Scholarly Information Access vs. Information Overload

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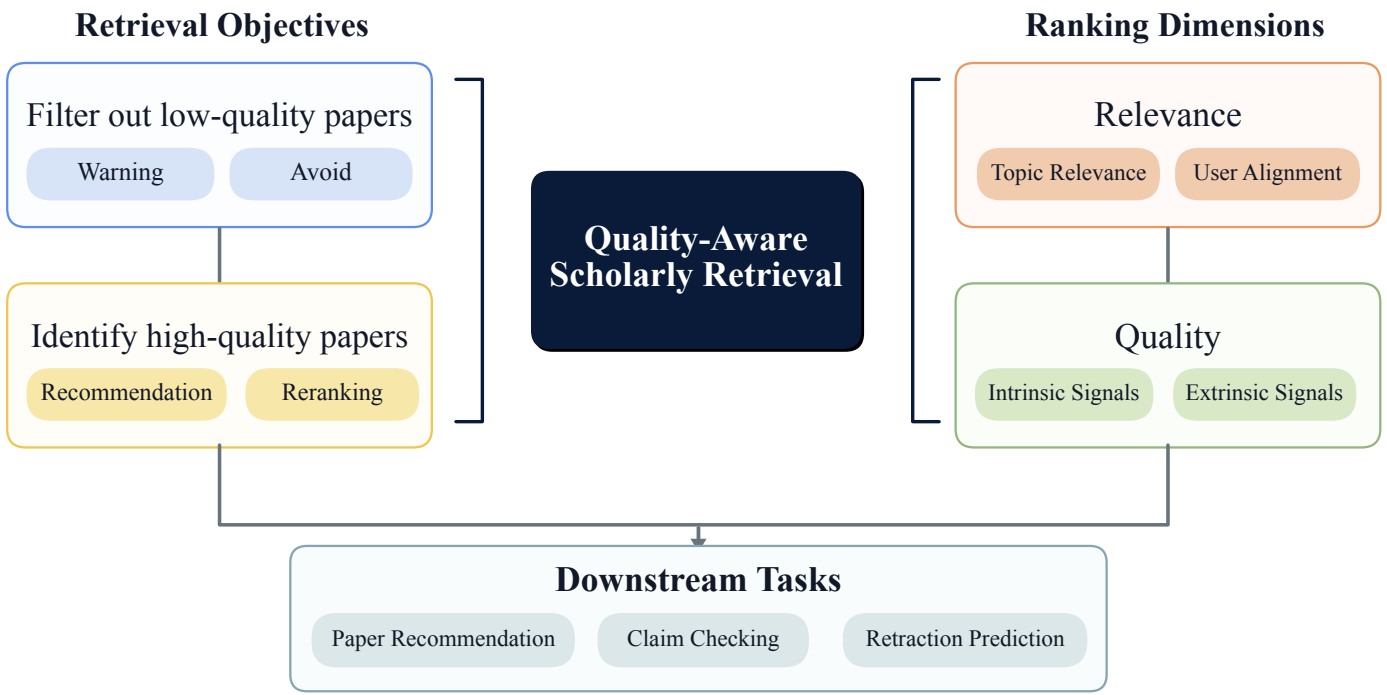
- Relevance
- Quality

The Way Forward – Challenges and Future Work

Conclusion

Quality-Aware Scholarly Information Access

- The challenge of information overload in scholarly communication requires effective IR methods to support researchers
- Relevance is not enough – two key aspects in scholarly information access:
  - (Topical) Relevance of scholarly documents to user queries and tasks
  - Quality of scholarly documents



Scholarly Information Access vs. Information Overload

Relevance

Bibliometric-enhanced Information Retrieval (BIR)

- Use of bibliometric and scientometric measures to enhance retrieval processes in scholarly search engines
- Examples:
  - Citation networks
  - Co-authorship networks
  - Journal Impact Factor
  - Altmetrics
  - Other scientometric indicators
- Specific IMRAD structure of scientific documents
  - Introduction
  - Methodology
  - Results
  - Analysis
  - Discussion

Ingo Frommholz, Guillaume Cabanac, Philipp Mayr, and Suzan Verberne. “Report on the 11th Bibliometric-enhanced Information Retrieval Workshop (BIR 2021)”. In: *SIGIR Forum* 55.1 (2021), pp. 1–9. DOI: 10.1145/3476415.3476426.

Bibliometric Indicators and Relevance

Coding scheme of the relevance evaluation behavior

Relevance criteria	Relevance factors	Relevance properties
quality	knowledge level	abstract
validity	content novelty	authors
accuracy	document novelty	journal
source reputation	source novelty	publication date
topic	understanding	title
depth	mental effort	language
scope	personal credence	snippet
recency	affectiveness	search terms
clarity	interface	altmetric score
type	rank	citations
method	time constraints	
theory	task constraints	
empirical	task difficulty	
accessibility		
affordability		
importance		

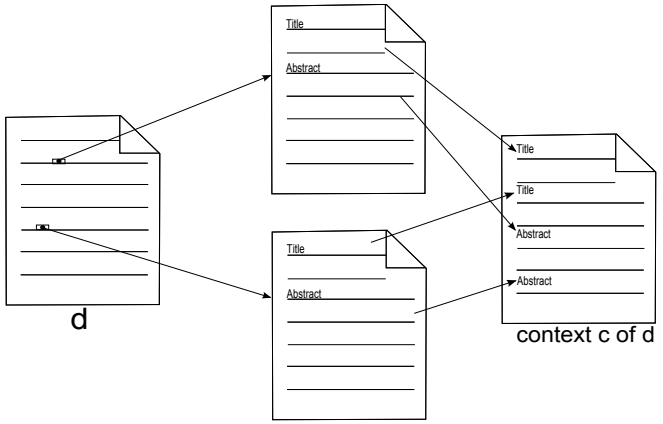
A user assesses *relevance criteria* by examining *relevance properties*, influenced by *relevance factors*.

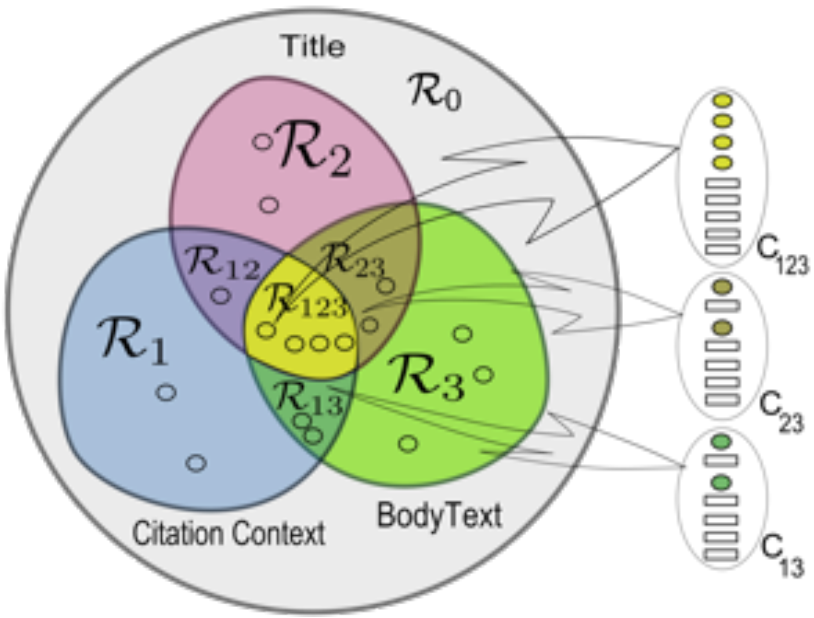
Jacqueline Sachse. "Bibliometric Indicators and Relevance Criteria - An Online Experiment". In: *Proceedings of the 11th International Workshop on Bibliometric-enhanced Information Retrieval (BIR 2021)*. 2021, pp. 69–77.

Bibliometric-enhanced Information Retrieval

Polyrepresentation, Clustering and Simulated Users

- Scenario: Bibliometrics and IR beyond ranked lists
- Probabilistic polyrepresentative clustering using citation contexts
- Simulated user – search strategy:
  1. User investigates total cognitive overlap cluster
  2. User jumps to different cluster based on preferences
  3. The user simulation creates a ranked list of documents





Muhammad Kamran Abbasi and Ingo Frommholz. "Cluster-based Polyrepresentation as Science Modelling Approach for Information Retrieval". In: *Scientometrics* 102.3 (2015), pp. 2301–2322. DOI: 10.1007/s11192-014-1478-1.

OMINO — LLM/RAG extension for GESIS Search

LLM-enhanced Retrieval and RAG

- LLMs and RAG to search the social sciences
- 1st prototype
  - AI-generated summaries of search results
  - Information scent based on extracted entities and topics



OMINO
Overcoming Multilevel INformation Overload

AI Summaries — do they work?

User Study

Research Questions

RQ1 How accurate are the AI-generated summaries in an academic search engine, and what are frequent error types?

RQ2 Can AI-generated search results summaries support social scientists during their search in an academic search engine?

- Research datasets ≈7,500 quantitative studies from 1945—2025) and ≈250,000 scholarly publications
- Collaboration with GESIS, Cologne, Germany – Kevin Schott, Dagmar Kern, Kanishka Silva, Daniel Hienert and Philipp Mayr

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Kevin Schott, Kanishka Silva, Ingo Frommholz, Philipp Mayr, Dagmar Kern, and Daniel Hienert. "AI Overviews in Academic Search: Evaluating AI-generated Summaries of Search Results in a Domain-specific Search Engine". In: *Proceedings of the Association for Information Science and Technology* 63.1 (2026). URL: http://arxiv.org/abs/2607.03421.

RQ1: Summary Accuracy

- Expert evaluation of 10 Summaries
- Labels: Correct, Partially Correct, Incorrect
- GPT-4o mini: 5 of the 10 summaries being labeled as “Correct,” 4 as “Partially Correct,” and 1 as “Incorrect.” In total, 9 out of 10 sample summaries were rated as either “Correct” or “Partially Correct.”
- LLama 4 Scout: 2 “Correct” and 8 “Partially Correct,” with no “Incorrect”
- Observed Error schemes (across models):
  - Inaccurate or Unsupported Content Descriptions
  - Unsubstantiated Guesses or Speculations
  - Omission of Search Results
  - Incorrect Study Identifiers
  - Inappropriate Grouping of Studies

RQ2: User Study

- Investigated whether AI-generated summaries (GPT-4o mini) influence user perceptions of workload, usefulness, satisfaction, and confidence during search
- 30 participants (social science background) performed search tasks with and without AI-generated summaries
- Lower mental demand with AI summaries, but no significant differences between the *AISummary* and *NoSummary* conditions (NASA–TLX scores or user-reported measures of usefulness, satisfaction, and confidence)
- Frustration scores showed a non-significant trend in favor of summaries, suggesting potential emotional benefits

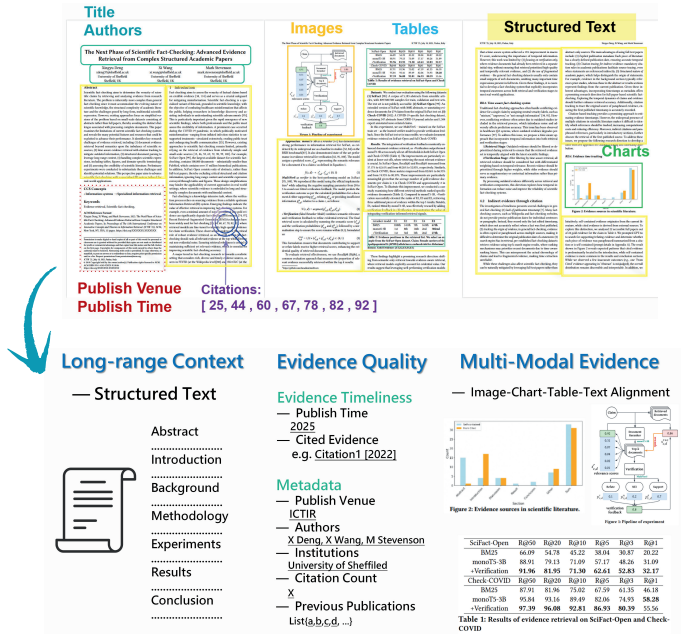
$\rightsquigarrow$ Summaries might reduce perceived mental effort without negatively affecting other subjective workload or satisfaction measures

Scholarly Information Access vs. Information Overload

Quality

Extrinsic and Intrinsic Quality Signals

- Extrinsic signals: external indicators of quality
  - Citations, peer review status, venue reputation, author reputation, temporal aspects (e.g., evolving evidence and outdated results), adherence to reporting guidelines
  - Not all available for new publications or preprints
- Intrinsic signals: internal content-based indicators
  - Internal coherence, multi-modal alignments, scientific rigour, replication status, data availability



Xingyu Deng, Xi Wang, and Mark Stevenson. "The Next Phase of Scientific Fact-Checking: Advanced Evidence Retrieval from Complex Structured Academic Papers". In: *Proceedings ACM ICTIR 2025*. ACM, July 2025, pp. 436–448. DOI: 10.1145/3731120.3744614.

Example: Evaluating Medical Text Quality with LLMs

- Task context: de-identified clinical letter generation
- BioGPT: assessing clinical information preservation via reference comparison
- GPT-3.5-Turbo: clinical soundness and narrative coherence assessment
- Limitations: coarse-grained judgments; limited supporting evidence
- Potentially transferable to research paper quality evaluation

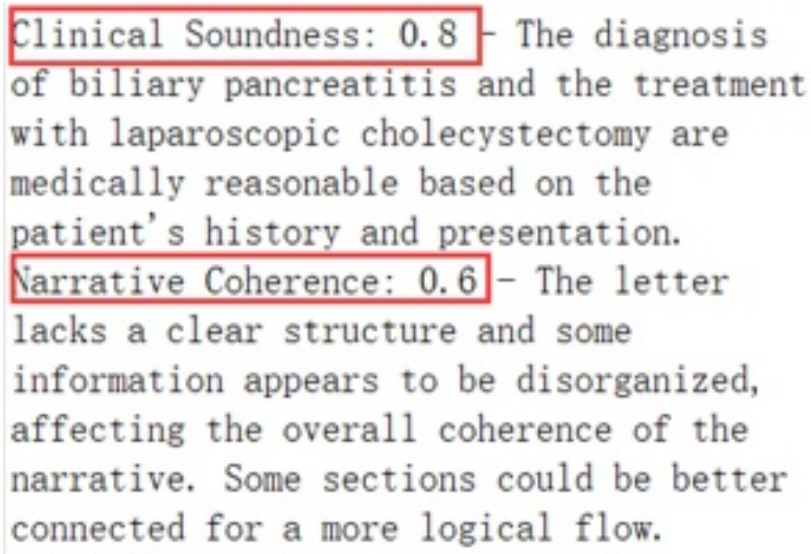


Figure: LLM-generated example of quality assessment

Libo Ren, Samuel Belkadi, Lifeng Han, Warren Del-Pinto, and Goran Nenadic. "Synthetic4Health: generating annotated synthetic clinical letters". In: *Frontiers in Digital Health* 7 (2025), p. 1497130.

REF 2021 Quality Levels

UK Research Excellence Framework (REF) 2021 defined quality based on three main criteria:

Originality "the extent to which the output makes an important and innovative contribution to understanding and knowledge in the field."

Significance "the extent to which the work has influenced, or has the capacity to influence, knowledge and scholarly thought, or the development and understanding of policy and/or practice."

Rigour "the extent to which the work demonstrates intellectual coherence and integrity, and adopts robust and appropriate concepts, analyses, sources, theories and/or methodologies"

Higher Funding Council of England. *Panel Criteria and Working Methods (2019/02) - REF 2021*. https://2021.ref.ac.uk/publications-and-reports/panel-criteria-and-working-methods-201902/index.html. (Visited on 08/05/2025).

REF 2021 Quality Ratings

The quality of research outputs is rated on a five-point scale:

- 4*: World-leading in terms of originality, significance and rigour
- 3*: Internationally excellent in terms of originality, significance and rigour but not world-leading
- 2*: Recognised internationally in terms of originality, significance and rigour
- 1*: Recognised nationally in terms of originality, significance and rigour
- Unclassified: Quality that falls below the standard of nationally recognised work or does not meet the published definition of research for the purposes of this assessment

Can ChatGPT assess research quality?

- Can we use LLMs as reliable estimators for research quality?
- Self-evaluation of 51 publications (accepted and rejected) using REF quality criteria
- Findings:
  - ChatGPT can provide consistent quality assessments
  - However, it tends to overestimate quality levels compared to human experts (among other limitations)

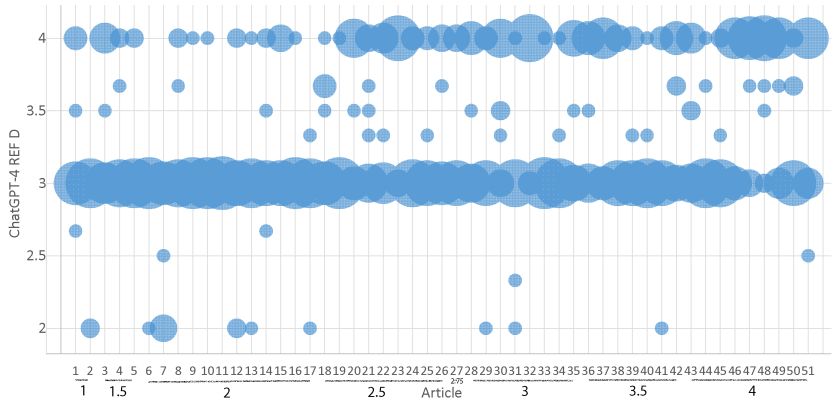


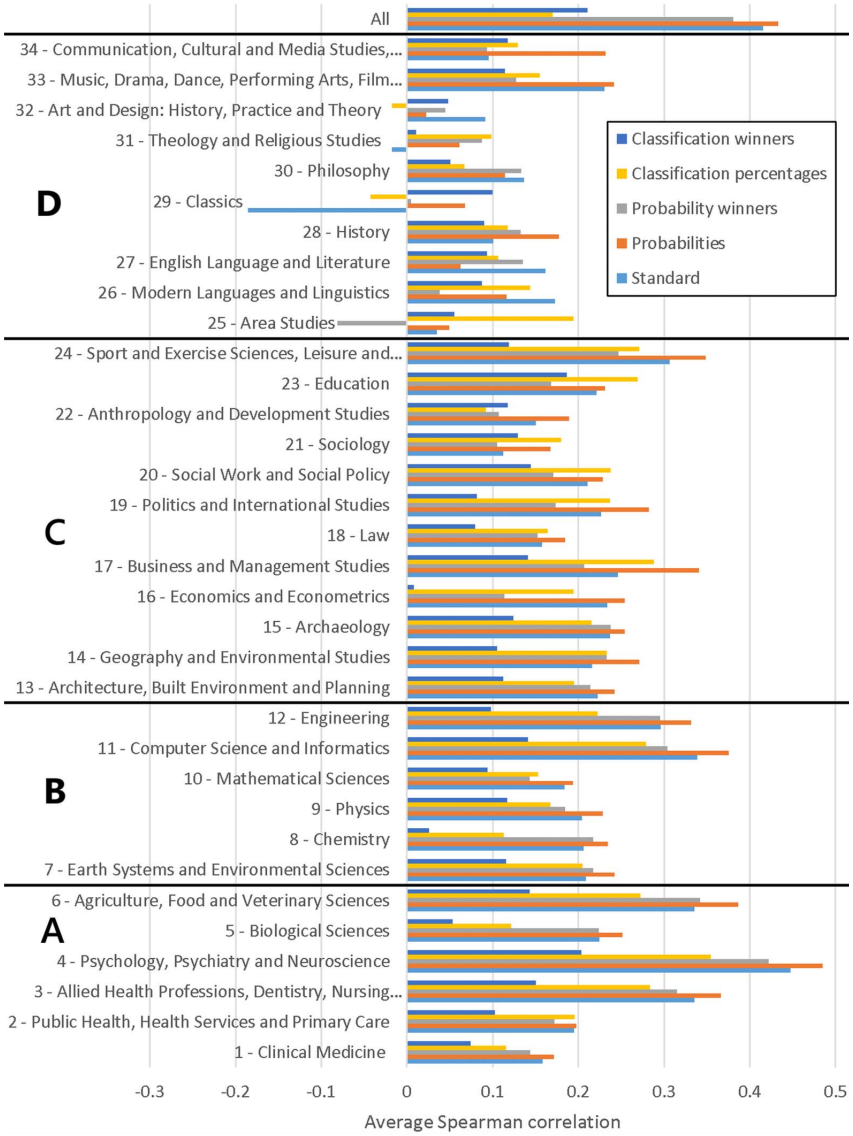
Figure 2. The range of REF star ratings given by the REF D GPT against the author's prior evaluation of the REF score of 51 of his open access articles. The area of each bubble is proportional to the number of times the y axis score was given by ChatGPT to the x axis article. My REF scores are marked on the x axis.

Mike Thelwall. "Can ChatGPT Evaluate Research Quality?" In: *Journal of Data and Information Science* 9.2 (Apr. 2024), pp. 1–21. DOI: 10.2478/jdis-2024-0013.

Mike Thelwall. *Evaluating Research Quality with Large Language Models: An Analysis of ChatGPT's Effectiveness with Different Settings and Inputs*. Aug. 2024. DOI: 10.48550/arXiv.2408.06752. arXiv: 2408.06752 [cs].

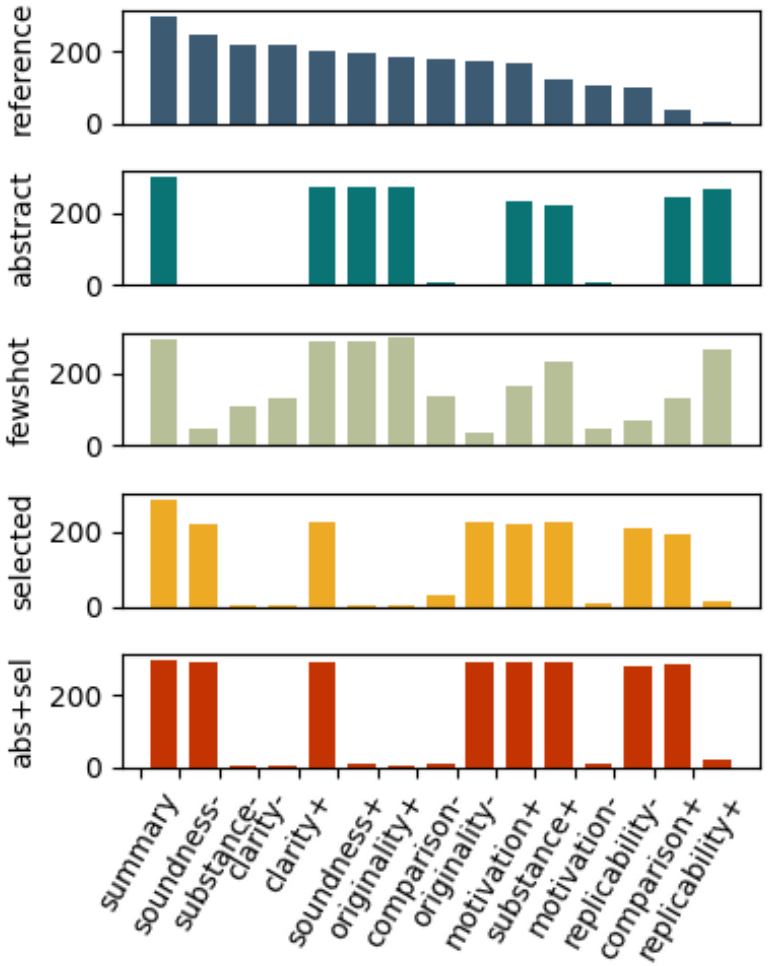
Research Quality Score Probabilities

- LLM quality scores improve when averaged over multiple (costly) iterations
- Score probability distributions can be exploited, fewer iterations required
  - Ask for percentages in prompt (classification percentages)
  - Token probabilities (leveraging ChatGPT's internal uncertainty)
- Slight improvements with token probabilities (on all 34 field-based REF Units of Assessment)



Mike Thelwall and Yunhan Yang. "Implicit and Explicit Research Quality Score Probabilities from ChatGPT". In: *Quantitative Science Studies* (2025), pp. 1–23. ISSN: 2641-3337. DOI: 10.1162/QSS.a.393.

LLM as Reviewer



- LLMs can generate scores based on reviews
- Feedback is often too positive
- LLMs have different aspect preferences
- LLMs not reliable reviewers (yet)

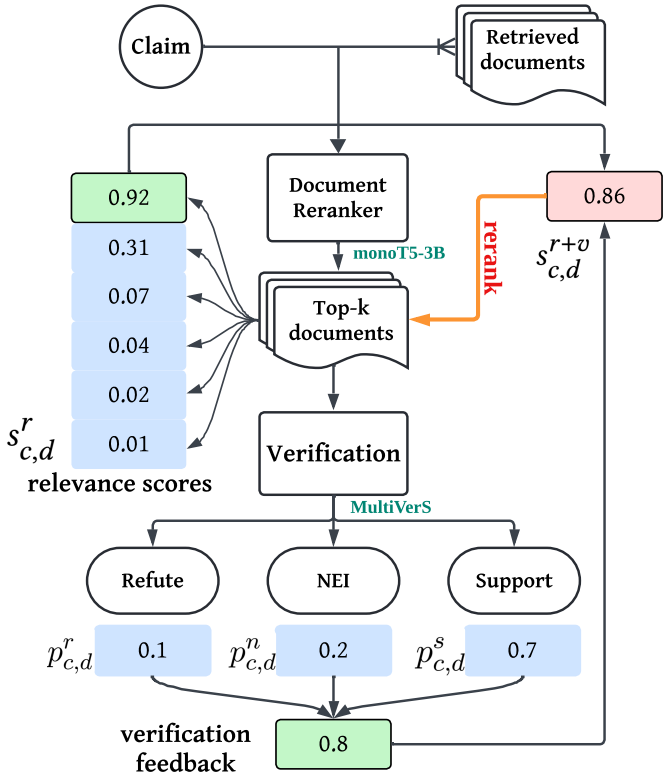
Ruiyang Zhou, Lu Chen, and Kai Yu. “Is LLM a Reliable Reviewer? A Comprehensive Evaluation of LLM on Automatic Paper Reviewing Tasks”. In: *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. Ed. by Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue. Torino, Italia: ELRA and ICCL, May 2024, pp. 9340–9351.

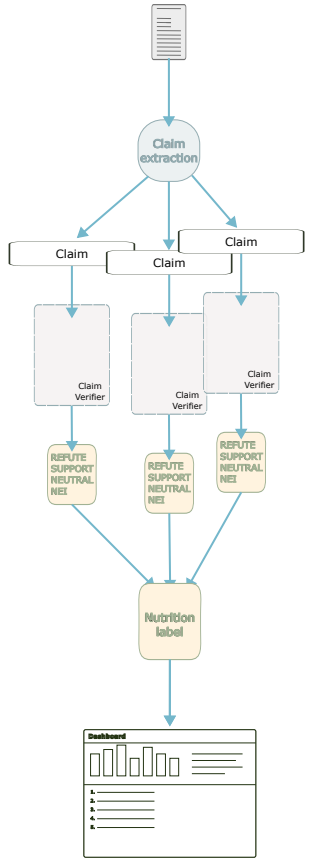
Using LLMs for Quality Assessment

- LLMs can assist in preliminary quality assessments of scholarly documents
  - They do not depend on citation data, which may be unavailable for new publications or preprints
- LLMs seem to be overly optimistic in their judgements (AI sycophancy?)
- Performance varies substantially by discipline.
- Not for serious peer review or hiring committees, but might still be useful for initial screening

Scientific Fact-Checking and Quality Assessment

- Automated verification of claims in or with scholarly documents
- In turn also requires qualitative evidence from scholarly documents





(from Deng et. al.)

Xingyu Deng, Xi Wang, and Mark Stevenson. "The Next Phase of Scientific Fact-Checking: Advanced Evidence Retrieval from Complex Structured Academic Papers". In: *Proceedings ACM ICTIR 2025*. ACM, July 2025, pp. 436–448. DOI: 10.1145/3731120.3744614.

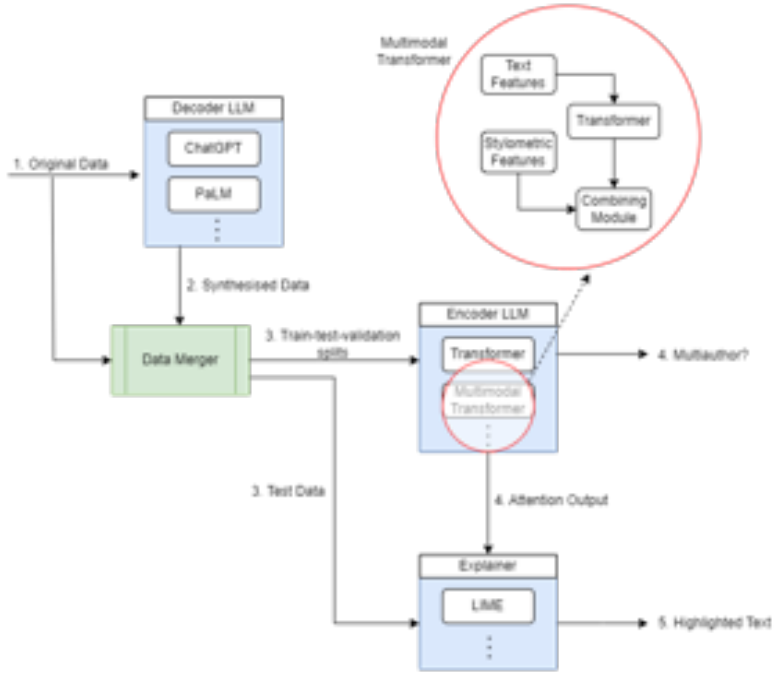
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Detecting AI-generated Publications

Authorship Attribution Example

- Scenario: Publications with automatically generated abstracts
- Multimodal transformer with stylometric and text features (BERT)
- Results prove the proposed model's effectiveness in detecting scientific articles with ChatGPT-aided abstracts

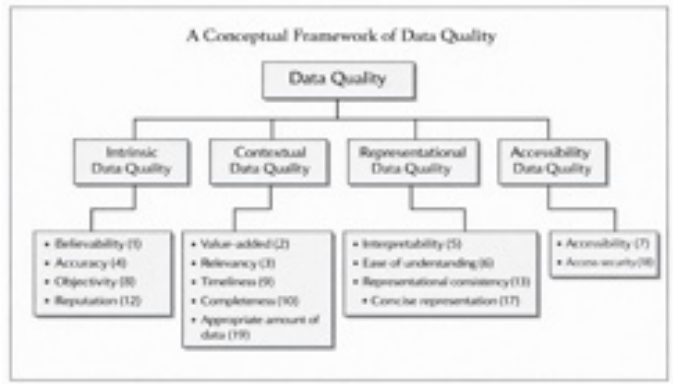


Model	Accuracy	F1	Precision	Recall
BERT [34]	0.55	0.52	0.57	0.56
Stylometric [35]	0.47	0.43	0.46	0.47
Character Ngram [9]	0.57	0.56	0.57	0.57
Word level TF-IDF [9]	0.52	0.51	0.52	0.52
MMT (BERT) + Stylometric (AWL + ASW + CFW)	0.98	0.97	0.98	0.98
MMT (BERT) + Stylometric (AWL + ASW + CFW) - CFW	0.94	0.94	0.94	0.94
MMT (BERT) + Stylometric (AWL + ASW + CFW) - ASW	0.94	0.94	0.94	0.94
MMT (BERT) + Stylometric (AWL + ASW + CFW) - AWL	0.94	0.94	0.94	0.94

Kanishka Silva and Ingo Frommholz. "What If ChatGPT Wrote the Abstract? - Explainable Multi-Authorship Attribution with a Data Augmentation Strategy". In: *Proceedings of the IACT - The 1st International Workshop on Implicit Author Characterization from Texts for Search and Retrieval (IACT@SIGIR 2023)*. Taipei, Taiwan: CEUR-WS.org, 2023, pp. 38–48. URL: https://ceur-ws.org/Vol-3477/paper6.pdf.

A Dual Two-Phase Framework for Data Quality Modeling

- Two-stage Survey
  - Initial survey to elicit data quality attributes from data consumers' perspectives
  - Collected attribute importance ratings and performed exploratory factor analysis (EFA) to extract underlying dimensions.



Richard Y Wang and Diane M Strong. "Beyond accuracy: What data quality means to data consumers". In: *Journal of management information systems* 12.4 (1996), pp. 5–33.

A Dual Two-Phase Framework for Data Quality Modeling

- Two-phase Sorting
  - Participants grouped intermediate-level items into 3–5 categories without predefined labels.
  - Category assignments were refined by removing low-importance and ambiguous dimensions, followed by participant confirmation.



Richard Y Wang and Diane M Strong. “Beyond accuracy: What data quality means to data consumers”. In: *Journal of management information systems* 12.4 (1996), pp. 5–33.

Summary: Scholarly Information Access vs. Information Overload

- Relevance and quality are both crucial for effective scholarly information access
- LLMs and AI can assist in quality assessment of documents and datasets (but seem to need close supervision)
- Scientific fact checking, detection of AI generated papers and data set quality important aspects for quality assessment

The Way Forward – Challenges and Future Work

What is Information Overload?

Information Overload in Academia – Quality vs. Quantity

Scholarly Information Access vs. Information Overload

The Way Forward – Challenges and Future Work

- Challenges and Possible Solutions

Conclusion

The Way Forward – Challenges and Future Work

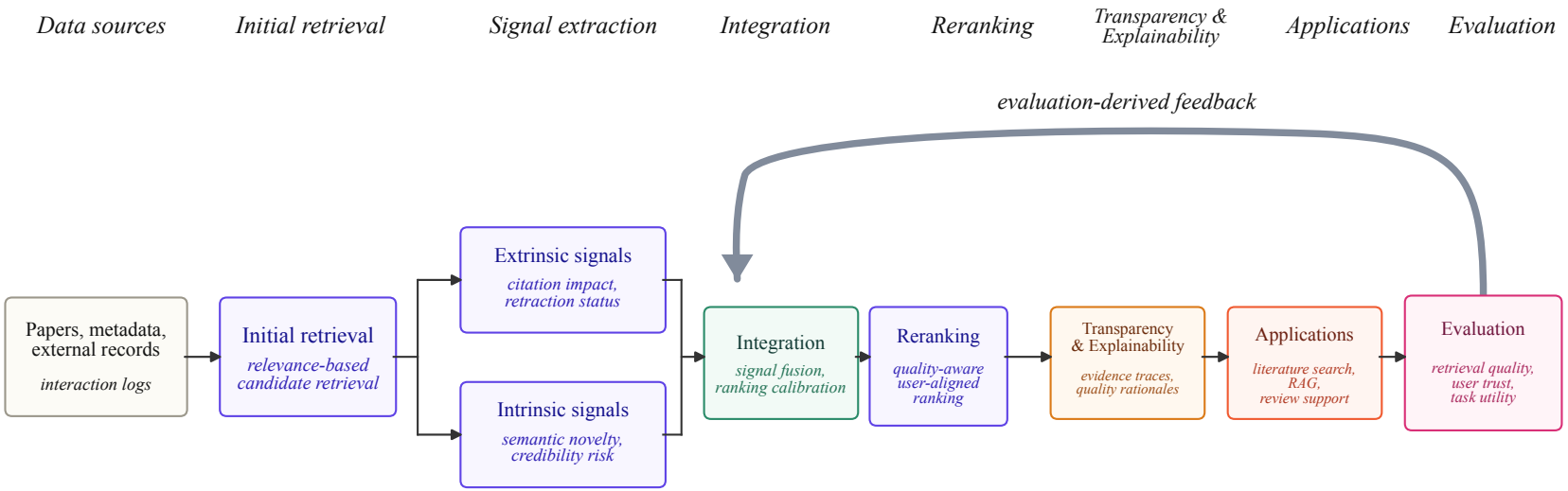
Challenges and Possible Solutions

Challenges for Academic Information Access Systems

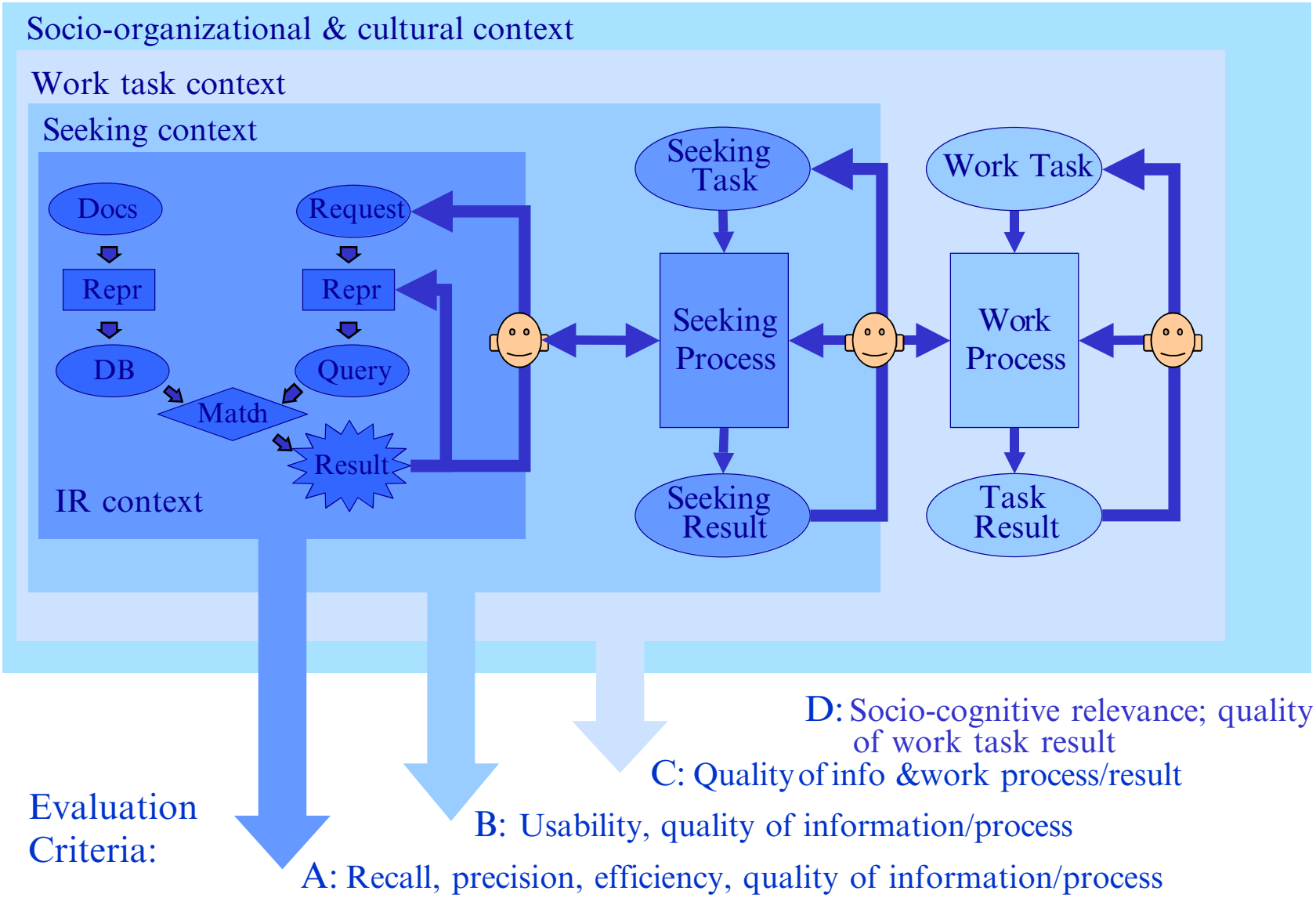
Desiderata

- Finding relevant publications not enough
  - Guide users by providing them with indicators for the quality and impact of publications
  - Transparency and explainability
- Support heterogeneous user groups (senior scholars, students, journalists, the interested public)
  - No “one size fits all”
  - Different disciplines and user groups have different demands
- Frictional interactive systems (Pickens [23]) with user in control and *machine in the loop*
- Multimodal scholarly document processing to scrutinise publications
- Tailored Datasets for evaluation (e.g., Badalova, Mayr & Sienkiewicz [3] Preprint2Paper)
- Evaluation initiatives, shared tasks

Quality-aware Retrieval Pipeline



A case for User Interaction in Context



Peter Ingwersen and Kalvero Järvelin. *The turn: integration of information seeking and retrieval in context*. Secaucus, NJ, USA: Springer-Verlag New York, Inc., 2005. ISBN: 140203850X.

A case for Information Seeking Strategies

ISS	Method		Goal		Mode		Resource	
	Scan	Search	Learn	Select	Recognize	Specify	Information	Meta-inform.
1	x		x		x		x	
2	x		x		x			x
3	x		x			x	x	
4	x		x			x		x
5	x			x	x		x	
6	x			x	x			x
7	x			x		x	x	
8	x			x		x		x
9		x	x		x		x	
10		x	x		x			x
11		x	x			x	x	
12		x	x			x		x
13		x		x	x		x	
14		x		x	x			x
15		x		x		x	x	
16		x		x		x		x

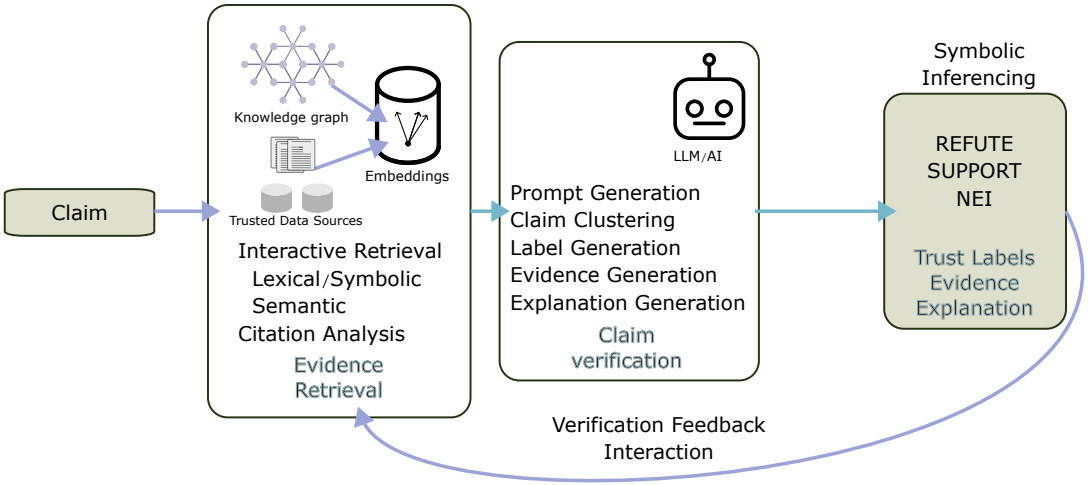
Nicholas J Belkin, Colleen Cool, Adelheit Stein, and Ulrich Thiel. "Cases, Scripts, and Information-Seeking Strategies: On the Design of Interactive Information Retrieval Systems". In: *Expert Systems with Applications* 9.3 (1995), pp. 379–395.

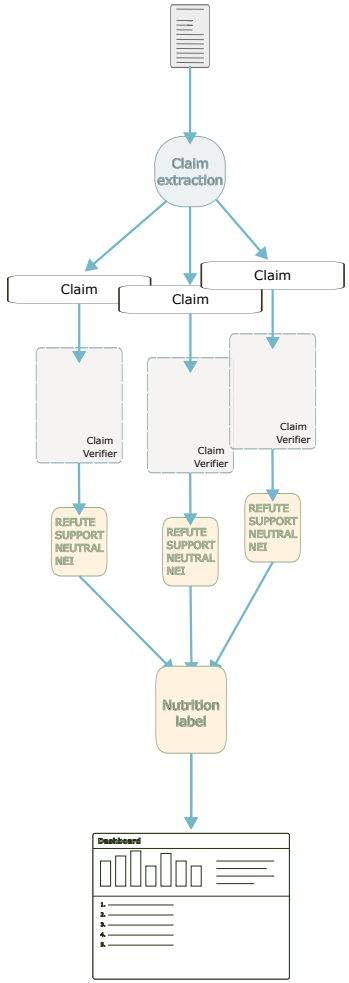
Chirag Shah and Emily M. Bender. "Situating Search". In: *Proceedings of the 2022 Conference on Human Information Interaction and Retrieval*. CHIIR '22. New York, NY, USA: Association for Computing Machinery, Mar. 2022, pp. 221–232. ISBN: 978-1-4503-9186-3. DOI: 10.1145/3498366.3505816.

Estimating Quality

- Need to define post-publication quality criteria (e.g., REF in the UK)
  - Different disciplines have different quality criteria
- Identify and integrate bibliometric/scientometric quality signals from scholarly documents
  - Citation-based signals (not available for new publications)
  - Altmetric signals
  - Content-based signals
  - Author-based signals
  - Venue-based signals
- Use of NLP and LLMs to extract intrinsic quality signals from scholarly documents

Estimating Quality: Interactive Claim Verification in Documents

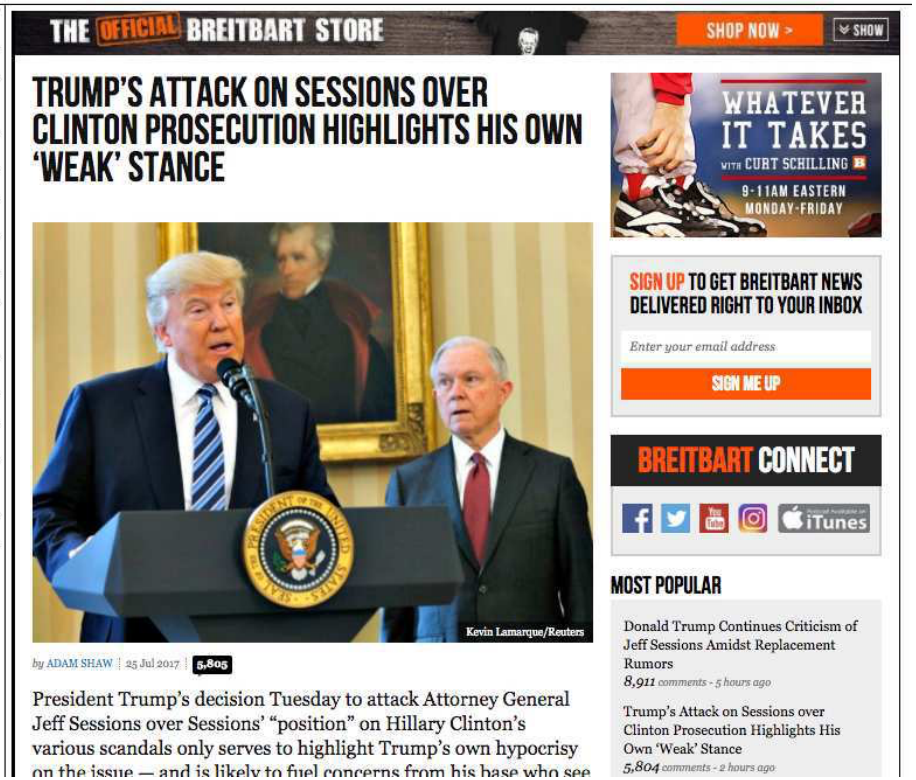




- Claim verification and internal consistency of a publication as a quality signal
- LLMs and verification feedback to extract and verify claims made in scholarly documents [12]
- Interaction and feedback loop allows users to guide the verification process

Reporting Quality and Other Measures – Information Nutritional Labels

INFORMATION NUTRITION LABEL		
Best Before: Jan 1, 2018		
Per 1000 words		**Recommended Daily Allowance**
Fact	30%	60 %
Opinion	40%	20 %
Controversy	9.0	--
Emotion	6.7	1.3
Topicality	8.7	5.0
Reading Level	4.0	8.0
Technicality	2.0	--
Authority	4.3	9.0
Viralness	--	1.0
Additional substances: advertising, subscription, invective, images (2), tweets, video clips		
Traces: product placement		



"It is not our intention to say what is true or what is false, right or wrong, and in particular not what is good or bad. That is, an Information Nutrition Label is not a substitute for a moral compass." (The authors of [17])

Norbert Fuhr, Anastasia Giachanou, Gregory Grefenstette, Iryna Gurevych, et al. "An Information Nutritional Label for Online Documents". In: *ACM SIGIR Forum* 51.3 (2018), pp. 46–66. ISSN: 0163-5840. DOI: 10.1145/3190580.3190588.

A Bit of Out-of-the-Box Thinking

- Many approaches (like claim verification) are RAG-based
- Often, the ‘R’ in RAG applies mainly lexical (BM25) or semantic search, possibly with neural re-ranker
- We have decades of IR research that can be relevant in this context!
- Two examples:
  1. Symbolic IR/AI with four-valued probabilistic logics
  2. User interaction and uncertainty with quantum probabilities

Conclusion

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- Information overload in academia is a significant challenge that requires innovative information access solutions

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- Information overload in academia is a significant challenge that requires innovative information access solutions
- We should estimate not just relevance, but also quality
  - Bibliometric-enhanced information retrieval (BIR)
  - LLMs and AI can assist (but only assist) in quality assessment
  - Scientific fact-checking might be able to support quality assessment

Conclusion

- Information overload in academia is a significant challenge that requires innovative information access solutions
- We should estimate not just relevance, but also quality
  - Bibliometric-enhanced information retrieval (BIR)
  - LLMs and AI can assist (but only assist) in quality assessment
  - Scientific fact-checking might be able to support quality assessment
- Think out-of-the-box!
  - Solutions should be user-centric, incorporating interactive IR and user feedback
  - Use the whole IR machinery and models


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